

Increasing extent of periods of no flow in intermittent waterways promotes heterotrophy

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SUMMARY

1. Current trends indicate that the duration of the period with no surface flow experienced by temporary waterways is increasing. Even though several studies have already characterised the effects of non-flow periods on stream ecosystems, the consequences of their increasing duration remain poorly understood.

2. Replicate artificial streams were used to characterise the effects of the duration of the non-flow period on stream biofilm processes. We hypothesised that increased duration of the non-flow period would differentially influence autotrophic and heterotrophic processes and promote heterotrophy (lower P : R ratios). We aimed to identify possible non-linearities and thresholds in the disturbance–response relationships between the duration of the non-flow period (six treatments) and stream biofilm processes (six ecological variables representing autotrophic or heterotrophic processes).

3. The duration of the non-flow period proved to be important for the balance between autotrophic and heterotrophic processes in stream biofilms, with the autotrophic processes being less resistant but more resilient. Autotrophic and heterotrophic processes also differed in terms of disturbance–response relationships, which were linear for most autotrophic ecological variables and sigmoid logistic for heterotrophic variable. Hence, non-flow periods had an immediate effect on autotrophic processes, but the effect on heterotrophic processes was delayed and completed only after crossing the ecological threshold of complete desiccation.

4. These differences involved a shift towards lower P : R ratios at longer durations of the non-flow period. However, the high resilience of autotrophic processes, mainly related to the ability of diatoms to recover, limited the shift towards lower P : R ratios to the non-flow period and to the first weeks after flow return.

5. This study demonstrates the complex effects of flow intermittency on stream biofilm processes, and emphasises the relevance of the duration of the non-flow period as a predictor of the effects on temporary waterways.

Keywords: artificial streams, ecosystem processes, global change, stream biofilm, temporary waterways

Introduction

Temporary streams and rivers, defined as waterways that cease to flow at some points in space and time along their course, are abundant in most biomes and are critical to the ecological health of river networks (Acuña *et al.*, 2014). These systems support high biodiversity and important ecosystem processes, as well as providing valuable goods and services. Recent work indicates that

69% of first-order streams (the smallest) below 60° latitude flow only intermittently, as do even a substantial fraction (~34%) of larger, fifth-order rivers (Raymond *et al.*, 2013). The spatial extent affected by non-flow events and the duration of these events within temporary waterways can vary from regular short dewatering and substratum drying of habitat patches to prolonged reach-scale events (Stanley, Fisher & Grimm, 1997). Temporary waterways are gaining increasing relevance as

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the number of waterways experiencing non-flow events increases due to direct (altered land-use patterns and water abstraction) and indirect (climate change) anthropogenic causes (Milly, Dunne & Vecchia, 2005; Döll & Zhang, 2010; Döll & Schmied, 2012).

Non-flow events alter the transport and processing of organic material and nutrients from adjacent ecosystems to rivers, as well as within the river itself (e.g. Acuña *et al.*, 2004; Amalfitano *et al.*, 2008). Consequently, shifts from flow to non-flow conditions (i.e. lack of surface flow leading to streambed desiccation) involve changes in biogeochemical cycles at the reach and basin scales (Acuña & Tockner, 2010; Corti & Datry, 2012). Understanding the effects of non-flow events on ecosystem processes in temporary waterways remains limited, because of the few studies that have explored the role of the duration of the non-flow period (e.g. Langhans & Tockner, 2006; Timoner *et al.*, 2012). However, all these studies have stressed the relevance of the duration of the non-flow period for ecosystem processes. Among the reported effects in the literature, the most remarkable is the uneven effect of exposure to non-flow conditions on autotrophs and heterotrophs. Non-flow events are known to severely impact biofilm algae in comparison to the biofilm bacterial community (Timoner *et al.*, 2014b), for which some degree of tolerance was observed in terms of both structure (Timoner *et al.*, 2014a) and function, as some extracellular enzyme activities remained during non-flow conditions, affecting the balance in C-N-P use by aquatic biofilms (Timoner *et al.*, 2012). Overall, the poor understanding of the effects of the duration of the non-flow period limits the ability to predict, for example, the balance between primary production and respiration at increasing durations of the non-flow period. Given the increasing number of waterways shifting from permanent to temporary, and the changes in the duration and spatial extent of the non-flow events, it is crucial to improve this predictive capability.

We evaluated the effects of the duration of the non-flow period on the functioning of stream biofilm heterotrophs and autotrophs by means of a manipulative experiment using artificial streams. We hypothesised that the duration of the non-flow period would differentially influence autotrophs (algae) and heterotrophs (bacteria), in the artificial streams, promoting heterotrophy because of a more severe reduction of autotrophs than of heterotrophs. The rationale behind this hypothesis was that autotroph habitat is restricted to the surface of rocks (epilithic biofilm) and sand (epipsammic biofilm), whereas heterotroph habitat also includes the subsurface

sediments, less prone to desiccation. Because of this, we expected autotroph structure and functioning to be impaired at all durations of the non-flow events, showing a linear disturbance–response relationship between the duration of the non-flow period and essential autotrophic functions (e.g. primary production). In contrast, we expected heterotrophic processes to be unimpaired at short durations of the non-flow events, but to be affected above a certain threshold (i.e. complete desiccation of surface and subsurface streambed sediments), and therefore to show nonlinear disturbance–response relationships between the duration of the non-flow period and their heterotrophic functions such as community respiration. Overall, we expected autotrophic processes to be less resistant than heterotrophic at short durations of the non-flow events and to show similar resistances at longer durations of the non-flow periods.

Methods

Experimental design

Effects of the duration of the non-flow period on biofilms were assessed in a series of artificial streams located at the indoor Experimental Streams Facility of the Catalan Institute for Water Research (Girona, Spain). Each artificial stream was assigned to one of six treatments following a randomised complete block design (6 treatments $\times$ 3 replicates per treatment = 18 artificial streams; distributed in three separate arrays of six artificial streams, with each treatment represented once per array). The control treatment experienced continuous flow throughout the experiment, whereas flow intermittency treatments experienced a period with no surface flow during 1, 3, 6, 12 or 20 days. Non-flow and consequent streambed desiccation was simulated by avoiding water flow in the artificial streams and by allowing the stream substrata to air-dry at room temperature ($\sim 22^\circ\text{C}$). The selected durations of the non-flow periods represented a trade-off between the observed natural ranges of non-flow duration and those that can be properly tested in the artificial streams. In fact, assessed non-flow period durations fall within the range for non-flow periods reported for temporary waterways such as chalk streams (Ledger *et al.*, 2008), but are shorter than those of Mediterranean temporary systems (81 ± 33 days in the Fuirosos stream; Butturini *et al.*, 2008). Flow was restored by the end of the non-flow periods and then continued for another 30 days. The impact of the non-flow conditions on ecosystem processes was assessed by comparing before and after the non-flow period, and

their recovery was assessed by analysing ecosystem processes during the 30 days after flow return. Furthermore, some processes were also analysed during the non-flow period to understand the impact of the non-flow period on ecosystem processes.

Experimental conditions

Physicochemical conditions were designed to mimic the conditions of natural streams. Each artificial stream was 2 m long, had a rectangular cross-section of 50 cm² and was set up as an open system with constant slope and steady, uniform flow. Artificial streams operated as once-through systems, with water in each individual replicate stream independent from other streams, and with a constant flow of 60 mL s⁻¹, resulting in mean water velocity of 0.88 ± 0.03 cm s⁻¹, and a water depth over the plane bed between 2.2 and 2.5 cm. Each artificial stream was filled with approximately 100 L of sand extracted from a nearby headwater Mediterranean stream (Fuirosos Stream) ($d_{50} = 0.70$ mm). The sand was sterilised with a Presoclave-II 30L autoclave (120 °C for 2 h) (JP Selecta S.A., Barcelona, Spain), and evenly distributed in the artificial streams to create a plane bed that facilitated the growth of the epipsammic biofilm. At complete water saturation, the porosity of the sand yielded a water content of 25% of the wet weight. Furthermore, 200 ceramic tiles (2.25 cm²), intended to mimic cobbles in natural streams, were positioned in two rows within each artificial stream above the sandy streambed to facilitate the growth of the epilithic biofilm.

Source water for the artificial streams was harvested rainwater, filtered through activated carbon filters. Prior to the experiment, the composition and temporal variability of the source water chemistry was assessed, and nutrients were added to approach the mean annual values of the Fuirosos stream (Butturini *et al.*, 2008). In fact, nutrients were added because of the relatively high concentration differences between the harvested rainwater and the Fuirosos stream. Thus, concentrations of phosphate and ammonium were raised to achieve the mean annual values of the Fuirosos streams (0.015 mg L⁻¹ P-PO₄³⁻ and 0.040 mg L⁻¹ N- NH₄⁺), by means of injection of concentrated solutions (KH₂PO₄ and NH₄Cl, respectively) and using a peristaltic pump (IPC pump; Glattbrugg, Ismatec, Switzerland). Daily cycles of photosynthetic active radiation (PAR) were defined as 14-h daylight + 10-h darkness. PAR was held constant at 150 µE m⁻² s⁻¹ during the daytime (natural conditions in the Fuirosos stream during spring; Acuña *et al.*, 2004). Water temperature was held constant at 20 °C by means

of a cryo-compact circulator (Julabo CF-31, Seelbach, Germany).

Background nutrient concentrations were assessed every 3–4 days from water collected from the end-channel area to assess the possible differences between the nominal and the observed concentrations of the injected nutrients (P-PO₄³⁻ and N- NH₄⁺). Differences were simply recorded, but injection rates of the concentrated solutions were not adjusted based on the differences. Water was filtered immediately through glass fibre filters (Whatman GF/F, Kent, U.K.) into pre-washed polyethylene containers, and transported to the laboratory. The concentration of P-PO₄³⁻ was determined colorimetrically (Alliance-AMS Smartchem 140; AMS, Frepillon, France). The concentrations of N-NO₃⁻ and N- NH₄⁺ were determined on a Dionex ICS-5000 ion chromatograph (Dionex Corporation, Sunnyvale, CA, U.S.A.). The dissolved organic carbon concentration (DOC) was measured on a Shimadzu TOC-V CSH coupled to a TNM-1 module (Shimadzu Corporation, Kyoto, Japan). Dissolved oxygen, pH and specific conductivity were measured daily by noon at each artificial stream using Wissenschaftlich-Technische Werkstätten (WTW) (Weilheim, Germany) hand-held probes to monitor these variables in the control treatment, as well as in the intermittency treatments when water was flowing. Water temperature in each artificial stream was recorded every 10 min during the entire experiment by means of VEMCO minilog (TR model; AMIRIX Systems Inc, Halifax, Canada) temperature data loggers (5–35 °C, ± 0.2 °C). Photosynthetically active radiation was also recorded every 10 min using 4 quantum sensors located across the whole array of streams (sensor LI-192SA; LiCOR Inc, Lincoln, NE, U.S.A.). Water content (separately in sand and ceramic tiles) was measured every 3–4 days during the non-flow period, calculated as the difference between wet and dry weight of the samples (including biofilm and substrata) after 24 h at 110 °C and expressed as the percentage of water of the wet weight.

Biofilm colonisation was allowed in the artificial streams for 5 weeks before the exposure to non-flow conditions. Biofilm was inoculated twice per week during the colonisation period using combined inocula from epilithic (growing on the surface of rocks) and epipsammic (growing on the surface of sand) biofilms of the Fuirosos Stream (Acuña *et al.*, 2004). During the colonisation period, biofilms were monitored twice per week for their maximum photosynthetic efficiency ($Y_{\max}$) and for extracellular activity of the enzyme alkaline phosphatase (APA). These measurements provided information on the physiological status of the biofilm (Sabater *et al.*,

2007) in the artificial streams, and were made to assess the physiological similarity of the biofilms in all the artificial streams before the exposure to non-flow conditions.

Ecological variables

The response of stream biofilms to non-flow conditions was assessed in terms of biomass [chlorophyll-*a* (Chl-*a*), ash-free dry mass (AFDM)], specific basal fluorescence (F_0) of primary producer groups (cyanobacteria, green algae, diatoms), photosynthetic efficiency ($Y_{\max}$), transformation of organic phosphorus (APA), and the overall biofilm metabolism as gross primary production (GPP) and community respiration (CR) (Table 1). These variables were measured at two overlapping frequencies: $Y_{\max}$ and APA were measured every 3–4 days during the entire experiment, including the non-flow period; AFDM, Chl-*a* and F_0 linked to the three major primary producers groups (cyanobacteria, green algae, diatoms), and GPP and CR were measured at the end of the colonisation phase, the day before implementing the non-flow conditions and at days 1, 4, 10 and 30 after the flow recovery (Table 1). The substrata analysed were ceramic tiles of 2.25 cm² in the case of the epilithic biofilm and sand samples obtained with a sample corer of 1.6 cm diameter in the case of the epipsammic biofilm (Table 1).

Chlorophyll-*a* concentration was used as a surrogate of algal biomass in the biofilm to evaluate the structural changes in the biofilm, and was measured after extrac-

tion in 90% acetone for 12 h in the dark at 4 °C (Steinman, Lamberti & Leavitt, 2006). Complete extraction of Chl-*a* was ensured with sonication (30 s, 360 W power, 50/60 Hz frequency, JP Selecta S.A.). After the filtration (Whatman GF/C, Kent, U.K.) of the extract, the Chl-*a* concentration was determined spectrophotometrically using a Lambda UV/VIS spectrophotometer (U-2000 Spectrophotometer; Hitachi, Tokyo, Japan) by following Jeffrey & Humphrey (1975). AFDM was used as an estimate of total biomass in the biofilm. For its determination, biofilm samples were dried at 105 °C to constant weight and later combusted at 450 °C for 4 h.

Maximum photosynthetic efficiency ($Y_{\max}$) was measured to evaluate the functional changes in the algal component of the biofilm. This parameter reflects the maximal efficiency of energy conversion at Photosystem II (PS II) reaction centres (Schreiber *et al.*, 2002), and is commonly used to evaluate the physiological state of algae and their response to different environmental stressors such as toxicants, light stress or desiccation (Sabater *et al.*, 2007; Corcoll *et al.*, 2012; Timoner *et al.*, 2012). $Y_{\max}$ was measured *in situ* with a portable pulse amplitude modulate fluorometer (Diving-PAM; WALZ, Effeltrich, Germany). The samples were set during 15 min under darkness, and a saturation pulse was performed to estimate the photosynthetic efficiency. The F_0 of dark-adapted cells was used as an estimation of relative algal biomass (Schmitt-Jansen & Altenburger, 2008). This estimation was based on the fluorescence recorded at four different wavelengths (470, 520, 645 and 665 nm) by means of Phyto-PAM fluorometer (version EDF; WALZ). In this fluorescence system, the four-channel fluorescence response is transformed to algal classes according to internal reference spectra of pure cultures (Walz, 1999). Fluorescence recorded at 470 nm is linked to the chlorophyll of green algae, whereas at 520 nm is related to that of diatoms, at 645 nm is mostly related to cyanobacteria and at 665 nm is related to the chlorophyll of the whole algal community (Walz, 1999; Jakob *et al.*, 2005). In this study, the fluorescence linked to each algal class (F_0 cyanobacteria, F_0 green algae and F_0 diatoms, respectively) was used as an estimation of their relative biomass.

Extracellular activity of APA is mainly associated with bacteria and cyanobacteria (Panosso, 2001), and it was measured to estimate the biofilm ability to transform organic phosphorus (Romaní & Sabater, 2000; Timoner *et al.*, 2012). APA involves the degradation of orthophosphoric monoesters into inorganic phosphorus (Klotz, 1992). APA was measured by means of a fluorescent-linked substrate (methylumbelliferyl, MUF). Colonised

Table 1 Ecological variables measured to assess the response in terms of biomass (Chl-*a*, AFDM), basal fluorescence of specific groups of primary producers (F_0), photosynthetic efficiency ($Y_{\max}$) and transformation of organic phosphorus (APA), as well as autotrophic (GPP) and heterotrophic processes (CR)

Code	Variable	Frequency (days)	Type
$Y_{\max}$	Maximum photosynthetic efficiency	3–4	<i>In situ</i>
APA	Alkaline phosphatase activity	3–4	Laboratory
F_0 cyanobacteria	Basal fluorescence of cyanobacteria	3–4	<i>In situ</i>
F_0 green algae	Basal fluorescence of green algae	3–4	<i>In situ</i>
F_0 diatoms	Basal fluorescence of diatoms	3–4	<i>In situ</i>
Chl- <i>a</i>	Chlorophyll- <i>a</i>	6–8	Laboratory
AFDM	Ash-free dry mass	6–8	Laboratory
GPP	Primary production	6–8	<i>In situ</i>
CR	Community respiration	6–8	<i>In situ</i>

tiles and sand samples were incubated for 1 h in the dark at 20 °C immediately after each sampling. Incubations were performed at a final concentration of 300 $\mu\text{mol L}^{-1}$ (Romaní *et al.*, 2004). Blanks and standards of MUF (0–100 $\mu\text{mol L}^{-1}$) were also incubated. At the end of the incubation, glycine buffer (pH = 10.4) was added (1 : 1, vol : vol), and the fluorescence was measured at 365 : 455 nm excitation: emission. Values were expressed as nmol MUF $\text{cm}^{-2} \text{h}^{-1}$.

Gross primary production and CR were measured to evaluate functional changes at the ecosystem level (i.e. epilithic and epipsammic biofilms), given that these metabolic rates are key processes for turnover of organic matter, inorganic materials and energy in a river. Their quantification is fundamental for understanding the mass and energy balances in ecosystems. Metabolic rates were assessed by changes in mean oxygen values in cylindrical recirculating chambers (Acuña *et al.*, 2008). Trays made of stainless steel wire mesh and containing 160 cm^3 of streambed sediments (epipsammic biofilms) and 36 cm^2 of ceramic tiles (epilithic biofilms) were used for metabolism measurements. These trays were located in the artificial streams during the entire experiment and the conditions within were equivalent to those experienced outside because of the mesh size used (diameter of 1 mm), so that epilithic and epipsammic biofilms developed within the trays under the same conditions. Thus, these trays were introduced into the chambers only during metabolism measurements and were later returned to the corresponding artificial stream. The chambers were made of acrylic glass (PMMA) (volume 0.96 L), and were provided with a submersible water circulation pump to avoid the existence of zones of low diffusion within the chamber. The incubations for each metabolism rate lasted for 2 h, and were carried out inside an incubator chamber (Radiber AGP-700-ESP, Barcelona, Spain) at the same temperature as in the artificial streams. GPP was measured under a PAR of $168 \pm 2 \mu\text{E m}^{-2} \text{s}^{-1}$, similar to the irradiance received at the artificial streams, and CR was measured in darkness. Dissolved oxygen concentration inside the chambers was measured continuously and logged at 30 s intervals with oxygen sensors (PreSens OXY-10mini, Regensburg, Germany). Metabolism rates were calculated according to Acuña *et al.* (2008).

Data analysis

Differences in biofilm variables between treatments were tested using analysis of variance (one-way ANOVA) before the exposure to non-flow conditions to check for

the uniformity of the initial biofilm community and at the end of the experiment (30 days after the flow recovery for each treatment) to check for the uniformity of the final biofilm community. The effects of the duration of the non-flow period on each variable were characterised by means of the resistance and the resilience (Holling, 1973; Grimm & Wissel, 1997; Uehlinger, 2000). Resistance was considered as an indicator of the capacity of minimal change, and was calculated as the % decrease experienced for a given variable between the last measurement before desiccation and the first measurement after flow return. Resilience was used as an indicator of the ability to recover rapidly after disturbance (Westman, 1978), and was calculated as the slope of the linear functions of the relationship between each of the response variables and the time since flow return (Grimm & Fisher, 1989), with all measurements after flow return included. Resistance and resilience were calculated with data from each artificial stream, so that 18 values were obtained. In the case of resilience, linear regression was used to determine the resilience values, but only those showing coefficients of determination (r^2) > 0.75 were accepted.

Disturbance–response relationships were characterised considering the duration of the non-flow event and the resistance values of each variable. Data were fitted to different expressions (linear, exponential decrease, polynomial of second and third order, sigmoid logistic of three and four parameters, and piecewise of two and three linear segments). Furthermore, the fitting of piecewise curves of two and three linear segments was used to identify ecological thresholds in the disturbance–response relationships (Groffman *et al.*, 2006; Brenden, Wang & Su, 2008). Model selection was based on the Akaike information criterion ($\text{AIC} = -2 \log\text{-likelihood} + 2 \text{ number of parameters}$) (Akaike, 1974; Johnson & Omland, 2004). AIC is an information theoretic optimality criterion that trades off bias (decreasing with the number of parameters in a model) with variance (increasing with the number of parameters in a model) to identify the most parsimonious model, which is the one with a minimal value of AIC. Models with the smallest AIC were considered best-fit, and were used to calculate the duration of the non-flow period causing a reduction of 50% from the before-exposure conditions. Pearson moment correlation analysis helped to identify the direction and strength of the relationships between variables. Normality of all variables was checked with the Kolmogorov–Smirnov test, and variables were log-transformed when necessary. All analyses were considered significant at $P < 0.05$ and were performed using

Table 2 Physical and chemical water characteristics of the artificial streams exposed to different durations of the non-flow period (0, 1, 3, 6, 12 and 20 days), averaged on a treatment basis. Artificial stream data were obtained from June to September 2012, whereas Fuirosos stream data were obtained from average values of several summers before non-flow events (Acuña *et al.*, 2004, 2005)

Treatment	Water temp (°C)	Dissolved O ₂ (mg L ⁻¹)	pH	Conductivity (µS cm ⁻¹)	N-NO ₃ ⁻ (mg L ⁻¹)	P-PO ₄ ³⁻ (mg L ⁻¹)	N-NH ₄ ⁺ (mg L ⁻¹)	DOC (mg L ⁻¹)
Fuirosos	23.50 ± 8.50	8.20 ± 5.50	7.9 ± 0.15	295 ± 51	3.5 ± 1.6	0.020 ± 0.007	0.040 ± 0.050	3.2 ± 0.65
0	21.22 ± 0.86	9.24 ± 0.37	8.40 ± 0.24	420 ± 23	1.95 ± 0.18	0.011 ± 0.005	0.026 ± 0.017	1.86 ± 0.16
1	21.58 ± 1.06	9.14 ± 0.36	8.35 ± 0.20	420 ± 20	1.90 ± 0.22	0.010 ± 0.004	0.029 ± 0.018	1.87 ± 0.22
3	21.39 ± 1.07	9.17 ± 0.37	8.38 ± 0.22	423 ± 20	1.93 ± 0.19	0.011 ± 0.004	0.030 ± 0.017	1.88 ± 0.20
6	21.51 ± 1.02	9.13 ± 0.38	8.34 ± 0.20	420 ± 21	1.90 ± 0.26	0.013 ± 0.004	0.029 ± 0.017	1.89 ± 0.22
12	21.33 ± 0.99	9.21 ± 0.40	8.41 ± 0.26	418 ± 20	1.89 ± 0.27	0.011 ± 0.004	0.030 ± 0.015	1.86 ± 0.19
20	21.30 ± 0.99	9.22 ± 0.34	8.40 ± 0.27	414 ± 18	1.92 ± 0.24	0.013 ± 0.004	0.034 ± 0.014	1.85 ± 0.19

DOC, dissolved organic carbon concentration.

SPSS (version 17.0, SPSS Inc., Chicago, IL, U.S.A.) or SigmaPlot (version 11.0, Systat Inc., Chicago, IL, U.S.A.).

Results

Experimental conditions

Temperature was held constant during the entire experiment and throughout the different treatments. Thus, air temperature in the Experimental Streams Facility (ESF) room averaged 22 °C, whereas water temperature in the

artificial streams averaged 21 °C in all treatments (Table 1). PAR cycles were also steady throughout the experiment, as well as the hydraulics and flow intermittency manipulation. Similarly, water chemistry was steady throughout the entire experiment and among treatments (Table 2): dissolved oxygen ranged between 9.13 and 9.24 mg L⁻¹, pH between 8.34 and 8.41, and specific conductivity between 414 and 423 µS cm⁻¹. Nutrient concentrations were slightly lower than the nominal concentrations (Table 2), probably because of the physical and biological processes within the artificial

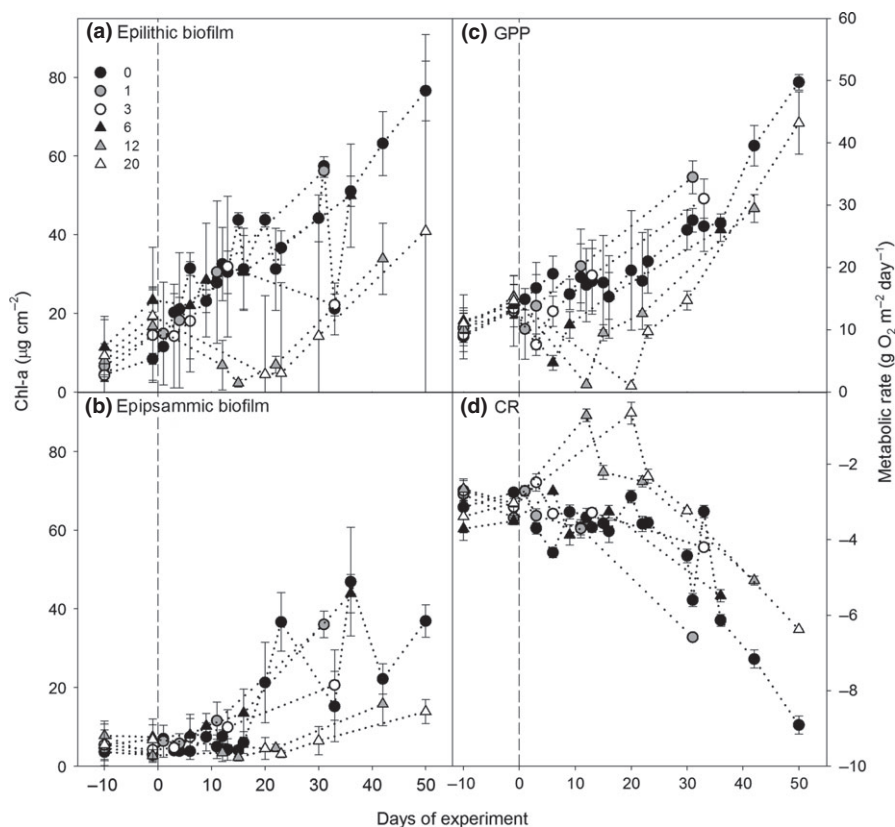


Fig. 1 Chl-*a* in the epilithic (a) and epipsammic (b) biofilms (µg cm⁻²), as well as gross primary production (GPP) (c) and community respiration (CR) (d) (gO₂ m⁻² d⁻¹). Note that symbols refer to the different treatments (durations of the non-flow period indicated with numbers) and that the vertical dashed lines indicate flow cessation in those treatments exposed to non-flow conditions.

streams. Water content before the exposure to non-flow conditions averaged 22.5% for the epilithic biofilm and 24.7% for the epipsammic biofilm. Water content did not change in the control treatment throughout the study period, whereas it decreased exponentially during all

the non-flow periods. As an example, the water content in the epilithic biofilm at the end of the non-flow period was 14% after 1 day under non-flow conditions, 5.3% after 3 days, 2.9% after 6 days, 0.11% after 12 days and 0.03% after 20 days.

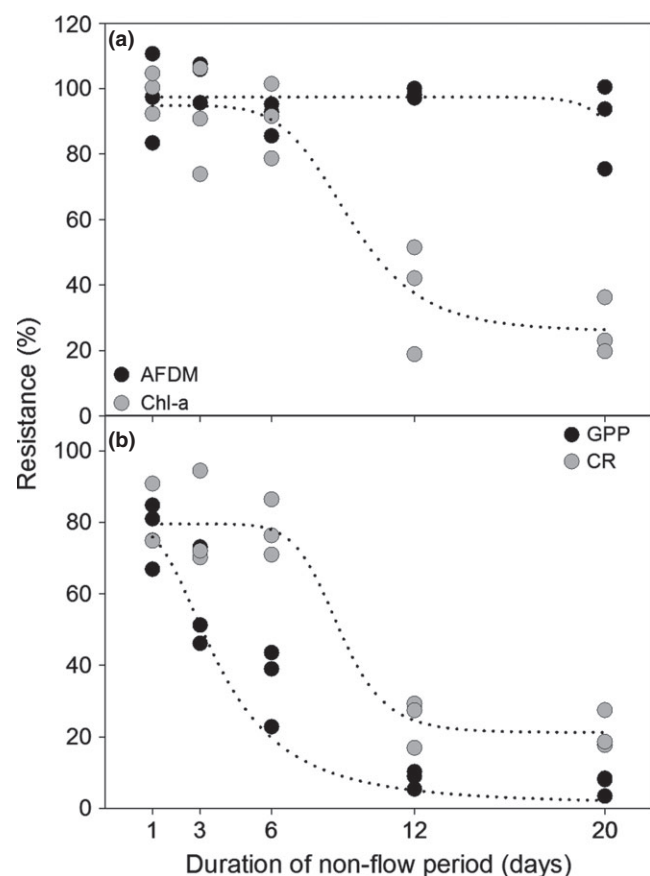


Fig. 2 Disturbance–response relationships between the duration of the non-flow period and ash-free dry mass (AFDM) and Chl-*a* (a), as well as gross primary production (GPP) and community respiration (CR) (b). Note that resistance values are given as percentages with respect to values before exposure to non-flow conditions.

Ecological variables

There were no significant differences in biofilm biomass between the artificial streams before the exposure to non-flow conditions (epilithic: $\text{ANOVA}_{\text{Chl-}a}$, $F = 0.99$, $P = 0.42$; $\text{ANOVA}_{\text{AFDM}}$, $F = 1.73$, $P = 0.14$). Initial chlorophyll-*a* was $15.8 \pm 11.6 \mu\text{g Chl-}a \text{ cm}^{-2}$ in the epilithic biofilm ($n = 54$) and $4.8 \pm 3.6 \mu\text{g Chl-}a \text{ cm}^{-2}$ in the epipsammic biofilm ($n = 54$), whereas AFDM was $3.4 \pm 0.7 \text{ mg cm}^{-2}$ in the epilithic biofilm ($n = 54$) and $5.2 \pm 2.1 \text{ mg cm}^{-2}$ in the epipsammic biofilm ($n = 54$). The temporal patterns of these two biomass variables (Chl-*a* and AFDM) were similar in the control treatment ($n = 20$, $r^2 = 0.74$). In contrast, non-flow decoupled their temporal patterns and Chl-*a* decreased to a greater extent than AFDM (Fig. 1a). In fact, the duration of the non-flow period involved significant differences between treatments at the end of the experiment for Chl-*a* in both epilithic ($\text{ANOVA}_{\text{Chl-}a}$, $F = 8.08$, $P < 0.05$) and epipsammic biofilms ($\text{ANOVA}_{\text{Chl-}a}$, $F = 19.62$, $P < 0.05$). For example, the Chl-*a* reduction after 6 days under non-flow conditions was 15%, whereas the reduction after 20 days was 75%. Differences were not as pronounced for AFDM, which showed significant differences in the epipsammic ($\text{ANOVA}_{\text{AFDM}}$, $F = 3.27$, $P < 0.05$), but not in the epilithic biofilm ($\text{ANOVA}_{\text{AFDM}}$, $F = 0.62$, $P = 0.64$). The different behaviour of these two biomass variables caused the duration of the non-flow period to be significantly correlated with the resistance of Chl-*a* in the epilithic and epipsammic biofilms ($n = 15$, $r^2 = 0.81$; $r^2 = 0.35$

Variable	Biofilm	Best-fit model	Goodness of fit (r^2)	50% (day)
AFDM	Epilithic	None	–	–
AFDM	Epipsammic	None	–	–
Chl- <i>a</i>	Epilithic	Logistic 4p	0.89	10.24
Chl- <i>a</i>	Epipsammic	None	–	–
Y_{max}	Epilithic	Logistic 3p	0.75	9.77
Y_{max}	Epipsammic	Logistic 4p	0.90	7.43
APA	Epilithic	Linear	0.47	–
APA	Epipsammic	None	–	–
GPP	Epilithic + epipsammic	Exponential 2p	0.90	2.84
CR	Epilithic + epipsammic	Logistic 4p	0.93	8.83

AFDM, ash-free dry mass; APA, alkaline phosphatase; CR, community respiration; GPP, gross primary production.

Table 3 Best-fit models of the relationship between duration of the non-flow period and the assessed ecological variables, as well as the duration of the non-flow period causing a reduction of 50% (in days)

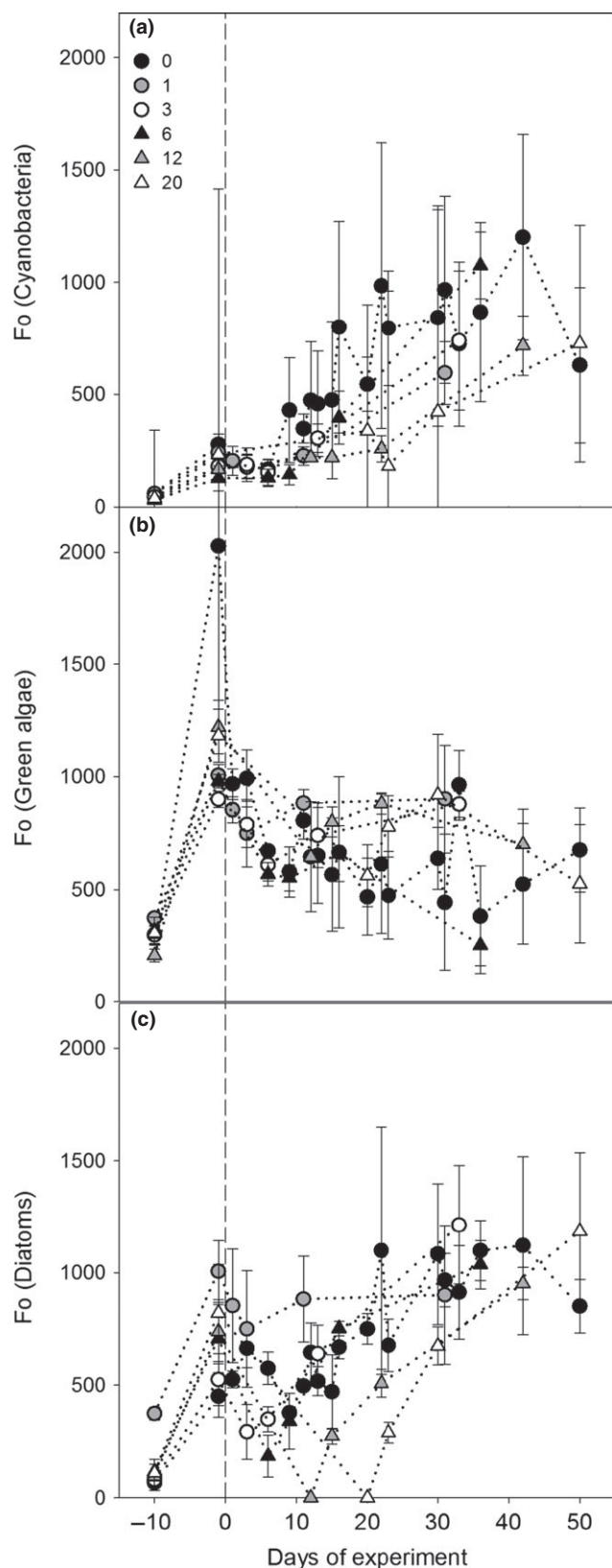


Fig. 3 Relative biomasses of the different algal groups during the experiment, as basal fluorescence (F_0) for cyanobacteria (a), green algae (b) and diatoms (c). Note that symbols refer to the different treatments (durations of the non-flow period indicated with numbers) and that the vertical dashed lines indicate flow cessation in those treatments exposed to non-flow conditions.

non-flow conditions was only *c.* 9.5%. Overall, these differences between Chl-*a* and AFDM were also reflected in the best-fit model of the disturbance–response relationships, as none of the considered curve types provided significant results for AFDM in either the epilithic or the epipsammic biofilms, whereas 89% of the variability of Chl-*a* in the epilithic biofilm was explained by the duration of the non-flow period (sigmoid logistic curve, four parameters; Table 3). In this case, the piecewise regression with three linear segments provided the same goodness of fit as the sigmoid logistic curve, indicating that an ecological threshold existed in the disturbance–response relationship for Chl-*a*.

There were different temporal patterns in the relative contribution of primary producers (cyanobacteria, green algae and diatoms) in the control treatment. While the contributions of cyanobacteria and diatoms increased over the course of the experiment, that of green algae decreased (Fig. 3). The different algal groups showed different disturbance–response relationships to the non-flow conditions. Cyanobacteria had an inverse relationship with duration of the non-flow period, and their relative contribution increased at longer durations ($n = 5$, $r^2 = 0.94$), whereas those of green algae and diatoms decreased. Green algae showed a sigmoid logistic relationship ($n = 5$, $r^2 = 0.74$) with an abrupt decrease after day 3 of exposure to non-flow conditions, while diatoms showed an exponential relationship ($n = 5$, $r^2 = 0.72$).

As for biomass, there were no significant differences in biofilm $Y_{\max}$ and APA between the artificial streams before the exposure to non-flow conditions (epilithic: ANOVA $_{Y_{\max}}$, $F = 0.85$; ANOVA $_{\text{APA}}$, $F = 1.27$). Maximum photosynthetic efficiency of the initial biofilm was 0.66 ± 0.04 in the epilithic biofilm ($n = 54$) and 0.65 ± 0.04 in the epipsammic biofilm ($n = 54$), whereas APA was 169 ± 34 nmol MUF $\text{cm}^{-2} \text{h}^{-1}$ in the epilithic biofilm ($n = 54$) and 150 ± 59 nmol MUF $\text{cm}^{-2} \text{h}^{-1}$ in the epipsammic biofilm ($n = 54$). The exposure to non-flow conditions decreased $Y_{\max}$, and the duration of the non-flow period was also affecting this parameter in the epilithic (ANOVA $_{Y_{\max}}$, $F = 4.61$, $P < 0.05$) and epipsammic biofilms (ANOVA $_{Y_{\max}}$, $F = 2.64$, $P < 0.05$). Similarly, there were significant relationships between duration of the non-flow period and resistance of $Y_{\max}$ in both the

respectively) but not to be related in the case of AFDM (Fig. 2a, Table 3). In fact, the decrease in AFDM was relatively low, as the reduction after 6 and 20 days under

epilithic ($n = 15$, $r^2 = 0.53$) and the epipsammic biofilms ($n = 15$, $r^2 = 0.77$); but this was not the case for APA in either biofilms. Thus, the duration of the non-flow period and APA were not inversely related as were the other biofilm variables, but directly related in both epilithic ($n = 15$, $r^2 = 0.47$) and epipsammic ($n = 15$, $r^2 = 0.32$) biofilms. In other words, exposure to non-flow conditions implied decreases in most assessed ecological variables, but increases in APA, and there were significant differences at the end of the experiment in both epilithic (ANOVA_{APA}, $F = 10.39$, $P < 0.05$) and epipsammic biofilms (ANOVA_{APA}, $F = 3.88$, $P < 0.05$). Differences between $Y_{\max}$ and APA were also reflected in the best-fit model of the disturbance–response relationship, as $Y_{\max}$ in both epilithic and epipsammic biofilms was explained when using sigmoid logistic curves, whereas this was not the case of APA (Table 3). Instead, none of the curve types considered provided significant fits for APA in the epipsammic biofilm, although APA in the epilithic biofilm showed a significant linear relationship with duration of the non-flow event (Table 3). In the case of $Y_{\max}$, the piecewise regression provided lower goodness of fit than the sigmoid logistic curves, and there were no differences between the piecewise of two and three linear segments, indicating there were no evident thresholds in these disturbance–response relationships. These differences between $Y_{\max}$ and APA were also reflected in the dynamics of these variables during the non-flow period, as $Y_{\max}$ decreased during the non-flow period in both biofilms (from 0.66 to 0.024 in the epilithic biofilm and from 0.65 to 0.026 in the epipsammic biofilm), but APA increased (from 169 to 400 nmol MUF cm⁻² h⁻¹ in the epilithic biofilm and from 150 to 312 nmol MUF cm⁻² h⁻¹ in the epipsammic biofilm) (Fig. 4).

Similarly to what was observed for the biofilm variables, there were no significant differences in the assessed metabolic rates between the artificial streams before the exposure to non-flow conditions (ANOVA_{GPP}, $F = 0.35$; ANOVA_{CR}, $F = 0.84$). The consistent similar response of all assessed variables was a guarantee of a similar functional status before flow intermittency was implemented. Initial GPP was 14.0 ± 3.1 mg O₂ m⁻² min⁻¹ ($n = 18$), whereas CR was -3.2 ± 0.5 mg O₂ m⁻² min⁻¹ ($n = 18$) (Fig. 1c–d). These two metabolic rates were similar along the experiment in the control treatment, where the P : R ratio was relatively constant (5.13 ± 1.08 , $n = 20$). CR rates in the control treatment were significantly related to Chl-*a* ($n = 20$, $r^2 = 0.70$) and AFDM ($n = 20$, $r^2 = 0.57$) in the epilithic biofilm, but not in the epipsammic biofilm. The GPP rates in the epilithic biofilm were also significantly related to Chl-*a* ($n = 20$,

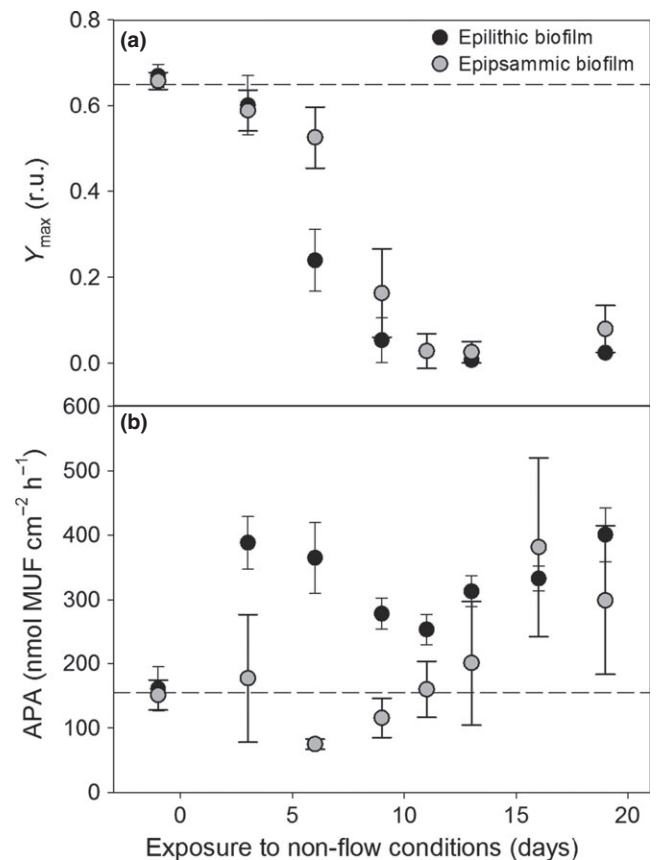


Fig. 4 $Y_{\max}$ (a) and alkaline phosphatase (APA) (b) during exposure to non-flow conditions in the artificial streams of the treatment with 20 days under non-flow conditions ($n = 3$). Note that black and grey dots refer to epilithic and epipsammic biofilms, respectively, and that dotted horizontal lines refer to the initial values of each variable before exposure to non-flow conditions.

$r^2 = 0.70$) and to AFDM ($n = 20$, $r^2 = 0.69$), but these relationships were not significant in the epipsammic biofilm. The GPP rates in the control treatment were significantly related to the F_0 of cyanobacteria and diatoms ($n = 20$, $r^2 = 0.33$ and 0.43 , respectively), but not to that of green algae. Overall, CR and GPP rates were related to both biomass indicators only in the epilithic biofilm, whereas this was not the case of the epipsammic biofilm, indicating that the metabolic rates were mainly coupled with the biomass changes in the epilithic biofilm.

Exposure to non-flow conditions caused both GPP and CR to decrease, and the decreases experienced by these metabolic rates differed among treatments, so that there were significant differences at the end of the experiment (ANOVA_{GPP}, $F = 10.78$; ANOVA_{CR}, $F = 10.53$). The duration of the non-flow period was related to resistance of GPP ($n = 18$, $r^2 = 0.65$) and resistance of CR ($n = 18$, $r^2 = 0.77$) (Fig. 2b), as well as with

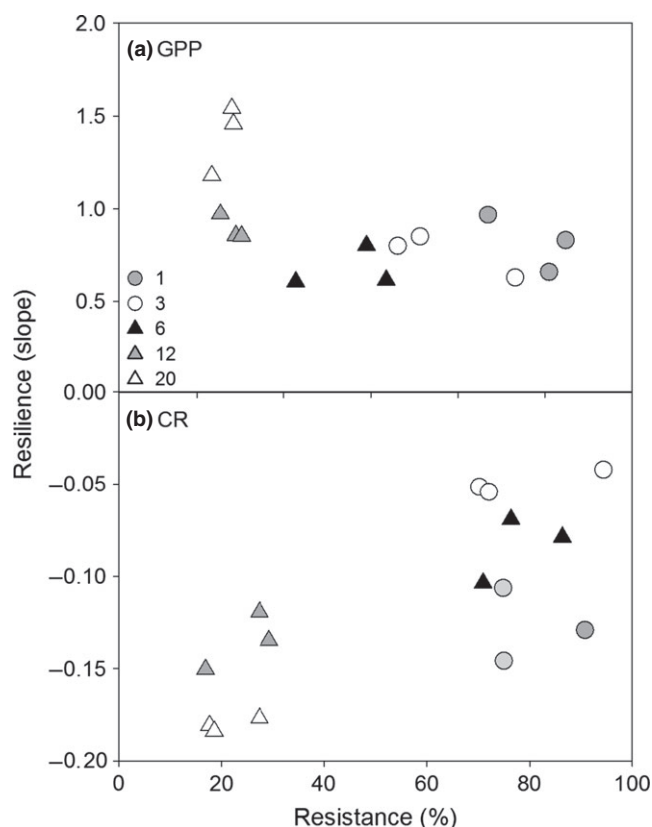


Fig. 5 Relationship between resistance and resilience for gross primary production (GPP) (a) and community respiration (CR) (b). Note that symbols refer to the different treatments (durations of the non-flow period indicated with numbers).

resilience of GPP ($n = 18$, $r^2 = 0.75$) and resilience of CR ($n = 18$, $r^2 = 0.70$). Specifically, the resistance of GPP was proportional to the duration of the non-flow period, decreasing from $77.5 \pm 9.4\%$ after 1 day of exposure to $8.1 \pm 2.5\%$ after 12 days (Fig. 5). Resistance did not decrease further, as there were no changes between 12 and 20 days of exposure to non-flow conditions, indicating that decreases below 8% are unlikely at longer durations. The maximum resilience in GPP was also observed at the longest exposure to non-flow conditions (Fig. 5). With regard to the relationship between resistance of CR and duration of the non-flow period, treatments of 1, 3 and 6 days of non-flow conditions showed resistance values around 80%, whereas resistance decreased to 25% at 12 and 20 days (Fig. 5). This abrupt change indicated the existence of a threshold between 6 and 12 days of duration of the non-flow period.

Differences existed regarding the best-fit models for the disturbance–response relationships of the metabolic rates. The GPP response was better described by an exponential curve with two parameters, while CR was better described by a sigmoid logistic curve with four parameters

(Table 3). The piecewise regression showed that GPP was better explained by a piecewise curve with two linear segments, whereas CR required a piecewise curve of three linear segments. This latter piecewise curve provided a goodness of fit equal to that of the sigmoid logistic curve with four parameters, a sign of its ecological threshold in the disturbance–response relationship. The P : R ratio decreased from 4 to 2 in the treatments with durations of the non-flow period longer than 6 days. However, this shift to a more heterotrophic metabolism only lasted 3–10 days after flow recovery, when the P : R ratios returned to pre-treatment levels. There were significant relationships between resistance of GPP and the resistance of F_0 of diatoms ($n = 5$, $r^2 = 0.98$) and of green algae to a minor extent ($n = 5$, $r^2 = 0.86$), but not of cyanobacteria owing to the previously mentioned inverse relationship with duration of the non-flow event.

Discussion

Most of the variables measured in the biofilm decreased accordingly to the duration of the non-flow period. However, the type of response was not uniform: some decreased at all assessed durations of the non-flow period; others did not change at short non-flow durations, but experienced abrupt changes later on. The variables responding at all durations of the non-flow period were apparently affected by the absence of surface water; those with later response (not before than 5–8 days) were not significantly affected by the absence of surface water but rather by the complete dewatering of the substratum, which occurred after day 6. Consistent with our hypothesis, those variables responding at all non-flow durations were mostly related to autotrophic processes (the effect on chlorophyll-*a* being the most obvious), whereas those responding only after crossing the ecological threshold of complete desiccation were related to heterotrophic processes. The GPP and CR patterns showed that shorter periods of non-flow might promote heterotrophy in biofilms of temporary watercourses. For example, a 50% reduction from pre-exposure conditions was achieved in 2.8 days for GPP, but 8.8 days were necessary to achieve the same reduction for CR. Similar patterns at the ecosystem scale were observed in the Fuirosos stream, where GPP decreased during the drying period, but CR did not change until the stream underwent watercourse fragmentation (Acuña *et al.*, 2005). In the artificial streams, the differential response resulted in a decrease of the P : R ratio at short durations of the non-flow period, integrative evidence of the shift towards heterotrophy.

Disturbance–response relationships might differ among ecosystem responses within the same ecosystem (Westman, 1978), and usually be expressed by differences in timing of ecological thresholds (e.g. Qian, King & Richardson, 2003; Miltner, 2010; Clapcott *et al.*, 2012). Shifting balances between ecosystem processes at increasing durations of the non-flow period might imply changes in the biogeochemistry of temporary waterways, which might be reflected at the basin scale because of the effects to downstream permanent waterways (Acuña & Tockner, 2010; Corti & Datry, 2012). Less organic carbon might be exported downstream if the duration of the non-flow period increases because of the presumed increase in the degree of heterotrophy in temporary watercourses. However, a shift to heterotrophy is limited to the non-flow period and to the first weeks after flow return, because GPP quickly recovers and the P : R ratio returns to values similar to those before exposure to the non-flow period.

Effects on biofilm variables

Previous studies have also reported linear responses to duration of the non-flow period (Stanley, Fisher & Jones, 2004; Stromberg *et al.*, 2005). For example, Stromberg *et al.* (2005) reported a negative linear relationship between the duration of the non-flow period and cover and richness of streamside herbaceous vegetation in a study along a gradient of duration of the non-flow period in the San Pedro River. These authors also reported differences in the magnitude of the response between different plant functional groups. The temporal scale integrated in field studies (i.e. years to decades (Stanley *et al.*, 2004; Stromberg *et al.*, 2005) is much longer than the scale integrated in our artificial streams (i.e. 60 days), and differences in the shape of the disturbance–response relationships might appear exponential in the short term but turn linear in the long term. The relative response of the primary producers in the artificial streams varied among algal groups. Diatoms showed lower resistance values than green algae, and cyanobacteria seemed to remain un-impaired by the exposure to non-flow conditions. Well-developed desiccation mechanisms of cyanobacteria allow them to resist adverse conditions in a range of geographic settings (Romaní & Sabater, 1997; McGregor, Marshall & Thoms, 2006), while some filamentous green algae also show desiccation-tolerance mechanisms (Morison and Sheath 1985). Stanley *et al.* (2004) also found that disturbance–response relationships to non-flow conditions differed among primary producer groups, and there were also

differences in terms of rates of recovery, as diatoms showed greater resilience values than the other primary producers groups. High resilience has also been observed for diatoms in Australia in streams that had been flowing for only 10–14 days (Peterson & Boulton, 1999). These differences among algal groups might explain the decoupling between the recovery of Chl-*a* and $Y_{\max}$ with flow onset, as most of the recovery of $Y_{\max}$ might have been caused by the recovery of the diatom photosynthetic activity rather by that of the entire primary producer community. Similarly, the high resilience of GPP (see below) might have been caused by the response of diatoms, and to a lesser extent of green algae. Overall, diatoms showed lower resistance and high resilience, whereas cyanobacteria showed high resistance and lower resilience. Even though these differences between primary producer groups are relative and based on the chlorophyll fluorescence relative to each algal group rather than on cell counting, our results are consistent with field observations (Timoner *et al.*, 2012) that a non-flow period of 112 days caused a severe decrease in both Chl-*a* and $Y_{\max}$, but a faster recovery of $Y_{\max}$ after flow return. Timoner *et al.* (2014b) also observed that diatoms had the lowest and cyanobacteria the highest resistance.

The response of APA differed from the other biofilm variables in terms of its nature and magnitude. This enzymatic activity increased during the non-flow period, most likely because of the continuous production of the enzyme, or because turnover rates of APA slowed down (Steinweg, Dukes & Wallenstein, 2012). A decoupling between the enzyme and its substrate quite likely occurred during the non-flow period, as cell lysis in the biofilm, freeing enzymes to the environment, may be general (Amalfitano *et al.*, 2008; Zoppini & Marxsen, 2011; Burns *et al.*, 2013). Furthermore, enzyme preservation may occur for weeks after desiccation in dry soils or aquatic biofilms (Perez-Mateos, Busto & Rad, 1991), and once free in the environment may bind to inorganic and organic soil particles, as well as to active or inactive cells (Burns *et al.*, 2013). In such a manner, enzymes are already available when favourable conditions occur and may be instrumental in the fast recovery of biofilms after non-flow periods (Romaní & Sabater, 1997). Despite the relatively high enzyme activities observed in several studies during non-flow periods, Amalfitano *et al.* (2008) concluded that the occurrence of non-flow events could affect carbon cycling through the freshwater microbial compartment, by temporarily limiting microbial mineralisation and altering bacterial community structure. Likewise, Timoner *et al.* (2014a,b) observed that during

the non-flow period in the Fuirosos stream, some tolerant bacterial groups and opportunistic bacterial groups took advantage of the new environmental conditions, possibly maintaining the relatively high levels of extracellular enzymatic activities observed during the non-flow (Timoner *et al.*, 2012).

Ecosystem resistance and resilience

The resilience of some of the biofilm variables was proportional to the disturbance caused by the exposure to non-flow conditions, thereby describing an inverse relationship between resilience and resistance. However, this relationship was not equivalent in the autotrophic and heterotrophic processes, as GPP showed a rather exponential relationship and CR a linear one. This inverse relationship between resistance and resilience was already predicted by Holling (1973), who justified it as the product of the evolutionary history of the systems in the face of the range of random fluctuations they have experienced. Furthermore, this relationship follows that depicted by Gunderson (2000), who envisaged ecosystems existing in 'valleys of stability' where the depth of the valley represents the system resistance to disturbance and the steepness of the valley sides represents system resilience. In this scenario, the initial speed at which ecosystems would return to their initial state would parallel the steepness on the valley walls, meaning that a U-shaped valley would imply an elastic pattern such as the one we observed in the artificial streams. The rates of recovery quantified for several biofilm variables in our study are indicative of the reversibility of the system once the disturbance ceases. However, we cannot be sure that this pattern would apply to longer durations of non-flow periods, that other ecological thresholds might not apply or that longer durations of the non-flow period would imply a hysteresis pattern of recovery to pre-disturbance conditions (Groffman *et al.*, 2006).

Thus, research needs to be pursued to determine whether this inverse relationship between resistance and resilience applies equally in field conditions, where stressors other than desiccation (e.g. high temperatures and irradiances) co-occur during non-flow periods, thus potentially involving harsher conditions than those in our artificial streams. In contrast to the artificial stream set-up, field conditions may allow faster recolonisation after exposure to non-flow conditions because of the likely existence of refugia from non-flow events, such as the deep hyporheic zone, or large leaf packs (Robson *et al.*, 2008). Therefore, results from this artificial stream

experiment should be cautiously applied to real-world situations. Nevertheless, controlled experimental systems, such as the artificial streams, eliminate irrelevant variability between treatments and replicates, giving much greater statistical power to identify the effects of interest. In our study, and particularly for GPP, a differential response of the different algal groups seems to be the governing mechanism behind this pattern. Apparently, the larger the reduction, the greater is the opportunity in terms of space and energy for the remaining diatoms and green algae to develop. Similarly, in a manipulative experiment assessing the effects of flow intermittency frequency on algal patch dynamics, Ledger *et al.* (2008) reported that resilience partially mitigated the effects of non-flow events at low intermittency frequencies, but not at higher frequencies. At low time frequencies, biofilms could persist in the form of small isolated nodules several millimetres in diameter, increasing the space available for colonisation by other opportunistic algae. Actually, this might be a potential explanation for the strong relationship between resistance and resilience in our experiment, when biofilm crusts remaining at the end of the dry periods could perform similarly to the biofilms described by Ledger *et al.* (2008). The differences between autotrophic and heterotrophic processes in the resistance–resilience relationship are behind the faster recovery of autotrophic processes, and thus the short-term shift to heterotrophy after exposure to non-flow conditions.

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